CHE 313 Transport Phenomena III

8/10

Quiz #3 (11/11/25) **Name:** Quan Huynh

target comp

10.0 kmoles of a feed that is 0.80 mole fraction acetone and 0.20 mole fraction ethanol is batch distilled in a system with a still pot (an equilibrium contact), a column that acts as one equilibrium stage (total two equilibrium contacts), a total condenser, and reflux to the column. p = 1.0 atm, the reflux ratio is L/D = 1.0, CMO is valid, and $x_{W,final}$ = 0.20.

Stage fixed $\Rightarrow x_D$ change

$$L/V = \frac{L}{D+L} = \frac{L/D}{1+L/D} = \frac{1}{1+1} = \frac{1}{2} \Rightarrow \boxed{L/V = 1/2}$$

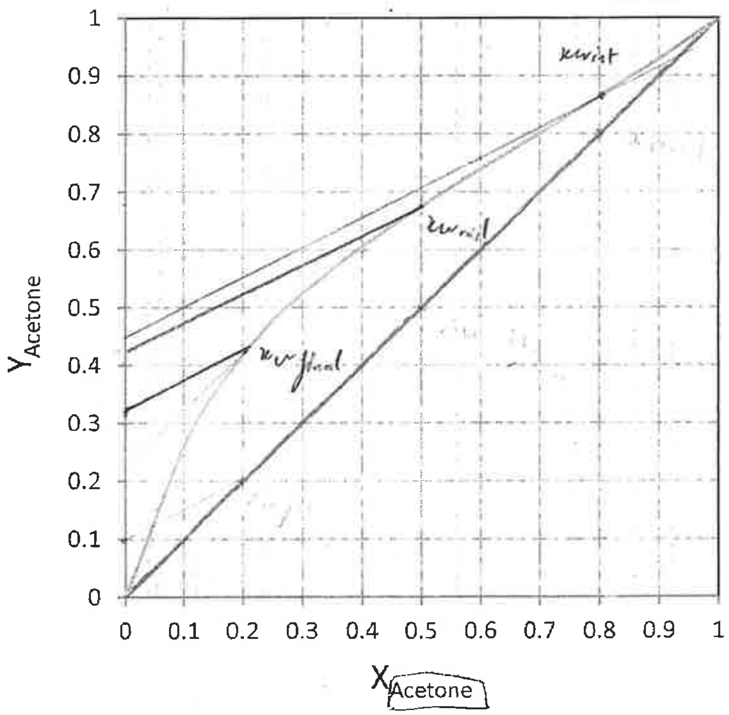
2

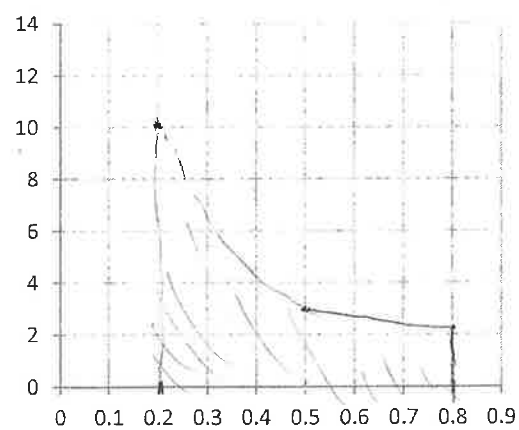
1) (2pts) Find the slope of operation equation, $y = (L/V)x + (1-L/V)x_D$

2) (4pts) Complete the table below using the VLE graph (draw OP lines and mark your x_w values on the graph), and mark points for area calculation.

3

	x_W	x_D	$x_D - x_W$	$1/(x_D - x_W)$
$x_{W,initial}$	0.8	0.9	0.1	10
$x_{W,middle}$	0.5	0.84 ~0.7	0.34	~4 ~~2.94~~
$x_{W,final}$	0.2	0.64 0.5	0.44	~3 2.27





$$Area = \frac{0.8-0.2}{6}\left(10 + 4 \times 2.94 + 2.27\right)$$
$$= \cancel{2.404} \text{ units}$$

3.23

3

3) (4pts) Find W_{final}, D_{total}, and $x_{D,avg}$.

$$W_{final} = F \exp\left(-\int_{x_f}^{x_F} \frac{dx_w}{x_D - x_w}\right) = F \exp(-\text{area}) = 10 \exp(-2.404) = \boxed{\cancel{0.904} \text{ kmol}}$$

0.4

$$D = F - W_{final} = 10 - F_{final} = 10 - 0.904 = \boxed{9.096 \text{ mol}}$$

$$x_{D avg} = \frac{F x_F - W_f x_f}{D} = \frac{10 \times 0.8 - 0.904 \times 0.2}{9.096} = \boxed{0.860}$$

Method is ok!

Simpson's rule: $Area = \frac{x_{Feed} - x_{final}}{6}\left[\frac{1}{x_D - x_W}\bigg|_{x_{final}} + \frac{4}{x_D - x_W}\bigg|_{x_{middle}} + \frac{1}{x_D - x_W}\bigg|_{x_{Feed}}\right]$

Rayleigh equation: $W_{final} = F\,exp\left(-\int_{x_{final}}^{x_{Feed}} \frac{dx_w}{(x_D - x_w)}\right) = F\,exp(-Area)$

$x_{D,avg} = (F*x_F - W_{final} * x_{final})/D_{total}$